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Assessing the benefits of high-speed broadband: Lessons from Australia's National Broadband Network (NBN)

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ABSTRACT

This study reviews the economic benefits analysis conducted in 2014 for the National Broadband Network (NBN), one of Australia's largest ever public infrastructure projects. The review identifies that the NBN's benefits were overstated due to error by an estimated $17–20 billion in 2014 present value terms, an amount sufficient to eliminate the NBN's forecast net economic benefits. To provide an updated view on the benefits of high-speed broadband in 2022, a stated choice experiment is conducted, finding that the WTP has grown since 2014 at an annual real rate of 8.8 percent for download speeds and 11.5 percent for upload speeds. Even at these growth rates, however, the benefits of the NBN remain materially lower than forecast in 2014. Further, the analysis finds high heterogeneity in the household WTP, with smaller and lower-income households reluctant to pay a premium for speeds much faster than 50 Mbps, indicating that the adoption of the fastest available broadband speeds is likely to remain low. The study highlights the potential value of cost-benefit analyses, but also the risks, particularly where they are relied on to justify enduring extensions of government ownership and control. The paper concludes by identifying learnings for policymakers and cost-benefit analysis practitioners seeking to assess and optimize the economic value of public infrastructure investments.

1. Introduction

The roll-out of the NBN has been one of Australia's most significant public infrastructure projects, requiring initial debt and equity funding of $49 billion from the Australian Government, as well as substantial structural and regulatory changes to the telecommunications industry. Although the decision to pursue the NBN was accompanied by forecasts of substantial net economic benefits, NBN Co's financial reports since inception tell a different story. At the end of the 2022 financial year NBN Co's cumulative losses were $32.7 billion.[1] Further, despite the network roll-out being completed in 2020, NBN Co continued to burn shareholder funds and cash, with an annual loss of $1.5 billion and net cash outflows of $813 million in 2022.[2] Whilst economic and accounting measures may differ, the losses raise questions regarding the accuracy of the forecast NBN economic benefits and the soundness of the government's investment decision.

This study considers three questions. First, were the benefit forecasts for the NBN robust at the time they were prepared in 2014? The study finds that the benefits were overstated by $17–20 billion in 2014 present value terms, an amount sufficient to eliminate the NBN's estimated net economic benefits. Second, how have the benefits of high-speed broadband and the NBN changed between 2014 and 2022? The study finds evidence of solid growth in the willingness to pay (WTP) for broadband speeds, including a real cumulative annual growth rate of 8.8 percent for download speeds and 11.5 percent for upload speeds. Despite these growth rates, however, the benefits of the NBN remain materially lower than forecasts in 2014, and high heterogeneity of demand indicates that the adoption of the NBN's fastest speeds may remain low. And third, what are the implications of the above findings for policy decision-makers? The study highlights the potential value of cost-benefits analyses but also the risks of estimation error, whilst identifying lessons for future broadband policy decision-makers and cost-benefit analysis practitioners.

In addressing the above questions, the paper focuses on the NBN economic cost-benefits analysis undertaken by the Vertigan Panel in 2014. The analysis was performed at a critical stage in the life of the NBN, when political and public interest were high, and the NBN's future strategic direction was being determined. Further, it remains the only

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[1] NBN Co, (2022), p.132.

[2] NBN Co, (2022), p.60 and p.130.

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substantive cost-benefit analysis of the NBN, sponsored by the government and containing original research. Fortunately, details of the methodology and the results were published at a reasonable level of granularity, allowing scrutiny of the findings and partial replication in 2022. Of note, it was argued by the Vertigan Panel that an advantage of cost-benefit analyses is that they promote accountability, by allowing comparisons between assessments at project commencement and outcomes at completion.[3] In relation to the NBN investment decision, this study starts the process of bringing home that advantage.

With similar broadband subscription rates across Australia, the United States, and Europe, the insights from this study are expected to the relevant across geographies.[4] Further, as governments globally announce ambitious plans to improve broadband infrastructure, an analysis of Australia's experience with the NBN is timely. Recent examples include the US government's allocation of $42 billion to support universal broadband access,[5] the UK government's commitment of £5 billion in funding support for gigabit-broadband access,[6] as well as the various country-based programs across the European Union.[7] An understanding of the changing household benefits of broadband speed, the heterogeneity of demand, and the potential risks in benefits estimation may inform policy choices.

The paper commences with a review of the related research, followed by background information on the NBN. The Vertigan Panel's benefit estimates from 2014 are then revisited and up-dated estimates generated for 2022, followed by a discussion of the research findings and implications.

2. Related literature

This study contributes to the stream of literature using WTP techniques to measure the benefits of fixed broadband. In an early study, Savage and Waldman (2005), using a discrete choice experiment to assess household WTP for alternative internet attributes, found the most important attributes to be speed and reliability. In a later study, Dutz et al. (2009), analysing demand during the transition from dial-up to broadband, used market data to determine that there had been a 60 percent increase in the US consumer surplus from broadband during the period 2005 to 2008 (an implied growth rate of 17 percent per annum). The study also identified, using survey data, a declining marginal WTP (MWTP) for faster speeds, with households willing to pay $28.60 per month to move from dial-up speeds to 5 Mbps, but only an extra $7.20 per month for additional speed increases to 50 Mbps. And Rosston et al. (2010), examining US household WTP for broadband in 2010 with a discrete choice experiment, found the consumer surplus per household had experienced a two- to three-fold increase since 2003 (an implied growth rate of 10–17 percent per annum). In addition, the study confirmed a declining MWTP for faster speeds, with households willing to pay $45.10 per month to move from slow to fast speeds, but only an extra $3.02 per month to move from fast to very fast speeds.

In an in-depth study, Liu et al. (2018) found the WTP for faster broadband speeds to be highly concave. Using a discrete choice experiment that allowed for a precise link between speed and WTP, the study reported that US households were willing to pay $2.34 per Mbps per month to increase speeds from 4 to 10 Mbps, declining to only $0.02 for increases from 100 to 1000 Mbps. In addition, Liu et al. (2018) identified that households with higher incomes and more education were willing to pay a premium for faster speeds, although this was not replicated in households that were urban, younger, or had children. Worden and Hambly (2022), assessing the WTP for speed improvements by households in Canada, found similar concave results, with households reporting an average MWTP of $2.39 per Mbps per month for download speeds at 50 Mbps, declining to $0.29 at 500 Mbps. And Lindlacher (2021), although not reporting on the WTP, examined the utility of fixed broadband in Austrian households in 2017. The study found the own-price elasticity of demand to be elastic, starting at −2.99 for speeds up to 20 Mbps and declining to −7.1 for speeds greater than 80 Mbps, indicating that the MWTP for the fastest speed plans was likely to be low.

As this paper uniquely considers the WTP for fixed broadband in the context of a public policy cost-benefit analysis, it also contributes to the stream of literature focused on cost-benefit analysis techniques. Using a variety of information sources, Flyvbjerg et al. (2002, 2009, 2021) has argued that most large projects experience cost overruns and benefits shortfalls, and that cost-benefit analysis is therefore broken, being susceptible to bad luck, optimism, and strategic misrepresentation. Other researchers, however, have questioned the validity of Flyvbjerg's findings, and argued that the jury is still out on the success of public projects and the role of cost-benefit analysis techniques (Love and Ahiaga-Dagbui, 2018; Volden and Welde, 2022). In this context, researchers have observed that robust ex post project evaluations, whilst often discussed, are rarely performed. Bellas and Kosnik (2019), for example, has argued that an extremely pertinent future research agenda would be to examine how well, or how poorly, ex ante cost-benefit analyses appear to do over time, providing insights for policy makers on their accuracy and learnings.

This study adds to the literature in three ways. First, it provides an updated analysis on the WTP benefits of fixed broadband in 2022, as well as new insights on the heterogeneity of demand. Prior to this study, the most recent comprehensive research conducted internationally into the WTP for fixed broadband relied on data from 2017 (Liu et al., 2018).[8] Since then, data volumes over fixed networks have increased, work and entertainment patterns have changed, and service providers have continued to offer faster speeds. Second, the study estimates the growth in the WTP from 2014 to 2022 and provides insights on expected future WTP growth trajectories. Whilst earlier researchers assessed the change in the WTP during the transition from dial-up to broadband speeds (Dutz et al., 2009; Rosston et al., 2010), this study is the first to assess the growth in WTP following the completion of the transition from dial-up internet. And third, unlike earlier research examining the WTP for broadband, this study examines the application of WTP techniques in a policy setting, where the results have been used to support a cost-benefit analysis and inform a public infrastructure investment decision.

3. Background to the original cost-benefits analysis

Since the early 1980s, the degree of competition in the Australian telecommunications market had gradually increased. By 2007, however, progress was mixed. Whilst competition was intense and dynamic in the mobile market, it had not developed to the same extent in the fixed line market. Further, for numerous reasons, such as the geographical spread of the Australian population, and the incumbency of Telstra, Australia's dominant telecommunications provider, there was a growing view that fixed line competition and infrastructure would develop only slowly (ACCC, 2010). During this period, broadband was increasingly being viewed as fundamental to economic and social development, but an area where there was growing concern that Australia was lagging the developed world.

At the 2007 Australian Federal election, broadband emerged as a significant political issue, with the Australian labour Party promising the construction of a national broadband network. Following the election,

[3] Vertigan Panel (2014a), p.52.
[4] OECD (2023), Table 1.2.1.
[5] Broadband USA, 2023.
[6] UK Parliament and House of Commons Library, 2023.
[7] For a summary see Progressive Policy Institute, 2022, p.12.

[8] The study by Worden and Hambly (2022) is distinguished on the basis that it was narrowly focused on the WTP for broadband *improvements*, testing a limited range of speed tiers using a contingent valuation survey technique.

however, implementation attempts were frustrated by a lack private sector willingness to invest in the government's proposed network model. In a dramatic response to this situation, in April 2009, the government announced the establishment of a new government owned business entity (NBN Co) to design, build and operate a high-speed national broadband network (NBN). The NBN was presented as a nation-building project that would improve access to high-speed broadband for all Australians, delivering download speeds of 100 megabits per second (Mbps) in urban areas using fibre-to-the-premise (FTTP) technology and speeds of at least 12 Mbps in remote areas with fixed wireless and satellite technologies.

From its inception, the NBN attracted divisive political debate and extensive media coverage.[9] Although pro-competitive due to its structurally separate, wholesale-only form, the NBN also introduced a monopoly, government-owned entity responsible for fixed-line network infrastructure, thereby reversing the historical trend of increased telecommunications infrastructure competition (Howell and Potgieter, 2020). Further, a key theme of the political and media focus became the costs and benefits of the NBN, and whether the NBN represented money well spent for taxpayers. Adding fire to the debate was a slow initial deployment of the NBN and a refusal by the government to perform a cost-benefit analysis, which became a focal point for criticism and questions regarding the legitimacy of the project (King, 2015).

In the lead-up to the September 2013 Australian Federal election, the opposition Federal Liberal/National Coalition party launched an alternative NBN deployment strategy, promising a faster and lower-cost rollout. This would be achieved by replacing the planned FTTP rollout with a multi-technology mix (MTM) rollout, involving a range of lower-cost access technologies with slower attainable broadband speeds.[10] In addition, as part of the strategy, a commitment was made to conduct an independent cost-benefit analysis of the NBN.[11] Following the election, the new government announced the terms of reference and a panel of experts to conduct the cost-benefit analysis (the Vertigan Panel), supported by government resources, commissioned research, professional consultancies, and an international advisory group.

The Vertigan Panel report, released in August 2014, concluded that there were substantial economic benefits associated with improving broadband speeds. It found that an unsubsidised strategy, where roll-out would only be extended to commercially feasible areas, would make Australians better-off by $24.0 billion in 2014 net present value terms. The MTM rollout strategy would generate net benefits of $17.9 billion, with lower net benefits due to the additional cost of subsidising fixed wireless and satellite access in remote areas. And the FTTP strategy would generate net benefits of just $1.8 billion, as the rollout would cost substantially more but without any material increase in benefits. The government endorsed the Vertigan Panel's cost-benefit findings, which both confirmed the benefits of high-speed broadband and provided support for the government's preferred MTM rollout strategy.

Several points are worth noting regarding the Vertigan Panel's benefits estimation approach. First, it focused exclusively on the benefits associated with improvements in broadband *speed*, considering other potential capability improvement to be less important.[12] Second, it concentrated on the *incremental* benefits of broadband speed improvements, by comparing the speeds that would become attainable under the NBN to the speeds that were available in 2014. And third, it calculated the benefits as the sum of the incremental amounts that households were estimated to be *willing to pay* for faster speeds, an approach consistent with a social cost-benefits analysis.[13]

To support a comprehensive benefits assessment, the Vertigan Panel applied three methodologies. The primary method, stated preference choice analysis, was considered the most robust and was the only method used by the Vertigan Panel to calculate the total benefits of the NBN. Using this method, household benefits were about $46 to $55 per month in the fixed line footprint (connecting urban area), and about half that amount in the non-fixed line footprint (connecting remote areas). The secondary methods, which were only used to generate alternative benefit estimates at the per household level, included the time savings method, which estimated benefits based on the expected time savings from faster broadband, and the actual take-up method, which estimated benefits by examining the actual price premiums being paid by those households already subscribing to faster broadband speeds in 2014. Using the secondary methods, household benefits were generally higher using the time savings method and lower using the actual take-up method (see Table 1).

4. Revisiting the original costs-benefits analysis

As with the Vertigan Panel, this study focuses on the choice modelling method. Further, the study concentrates on the benefit estimates for the fixed-line footprint, which the Vertigan Panel had forecast would comprise 93 percent of connected NBN premises. Note that in revisiting the Vertigan Panel's analysis, references to the Vertigan Panel encompass the multiple agents that supported the Vertigan Panel, and this study does not attempt to separately identify the specific agents that were assisting the Vertigan Panel.

4.1. Primary method – Choice modelling method

The Vertigan Panel's choice experiment was completed in May 2014, with 3312 survey respondents each performing eight choice tasks. The respondents were divided into two survey treatment groups: an "uninformed" group was given minimal information about what higher broadband speeds mean; and an "informed" group was shown information about activities that could be performed at different speeds. Overall household WTP was estimated by averaging the estimates for the two groups.[14]

Extracted from the Vertigan Panel report, Figs. 1 and 2 present the MWTP for download and upload speeds. Both figures report declining MWTP as speeds increase. For upload speeds, the MWTP for informed and uninformed respondents was similar (see Fig. 2). For download speeds, however, the informed respondents had a much higher MWTP than the uninformed respondents (see Fig. 1). At a download speed of 20 Mbps, for example, uninformed respondents had a MWTP of $0.50 per month for a 1 Mbps increase in speed (reading from Fig. 1), whilst informed respondents had a MWTP of $1.80 per month. The magnitude of this difference is surprising, with informed respondents willing to pay about three times more than uninformed respondents.

Fortuitously, the Vertigan Panel report contains sufficient attribute and formula data for Figs. 1 and 2 to be recreated (see details in Appendix A). The recreated figures are almost identical to the equivalent Vertigan Panel figures, with the exception that the download speed MWTP results for informed respondents are substantially lower (see Figs. 3 and 4). Returning to the earlier example, the recalculated estimate is that informed respondents have a MWTP of $0.69 for a 1 Mbps increase in speed from 20 Mbps, about 60 percent less than that the $1.80 presented in the Vertigan Panel's published chart.

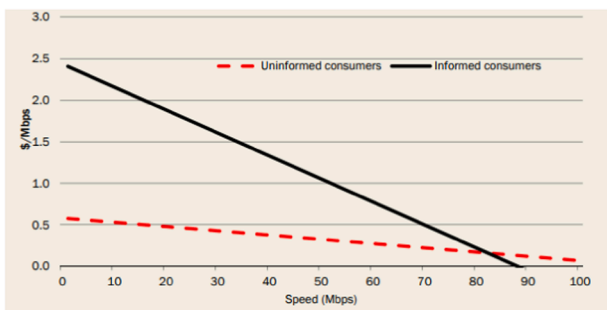
Adjusting for the above discrepancy, the Vertigan Panel's reported NBN incremental benefits per household are reduced by about 47 percent (see Table 2). Further, it can be demonstrated that this estimation error at the per household level cascades through to the Vertigan Panel's overall reported benefit results (see details in Appendix A).

[9] Over 55,000 media articles were published on the NBN over the period 2008-13. Source: Wilken et al. (2015), p.58.
[10] Liberal Party of Australia (2013), pp.5-6.
[11] Liberal Party of Australia (2013), p.12.
[12] Vertigan Panel (2014b), p.37.
[13] Vertigan Panel (2014b), p.24.
[14] Vertigan Panel (2014b), Appendix H.

Table 1
Vertigan Panel (2014) – Incremental NBN Benefits per Household per Month, 2014 Dollars.

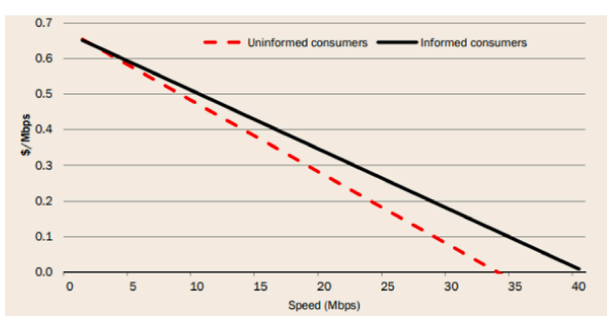
Footprint and Roll-out Strategy	Method: Choice Modelling			Method: Time Savings			Method: Actual Take-up[^^]		
	DL	UL	Total	DL	UL	Total	DL	UL	Total
Fixed Line Footprint									
Unsubsidised Rollout	$40.40	$6.10	$46.50	$43.61	$9.49	$53.10	–	–	$21.80
MTM Rollout	$40.50	$6.10	$46.60	$43.71	$9.50	$53.21	–	–	$21.80
FTTP Rollout	$45.80	$9.70	$55.50	$43.72	$9.51	$53.23	–	–	$26.30
Non-Fixed Line Footprint									
Unsubsidised Rollout	–	–	–	–	–	–	–	–	–
MTM Rollout	$23.00	$0.60	$23.60	$117.88	$12.94	$130.82	–	–	$16.70
FTTP Rollout	$23.00	$0.60	$23.60	$117.88	$12.94	$130.82	–	–	$16.70

Benefits incremental to the "No further roll-out" strategy. DL = download speed, UL = upload speed.
[^] Unsubsidised Rollout did not include rollout to non-fixed line footprint.
[^^] The actual take-up method did not breakdown WTP into DL and UL speeds.
Source: Recreated from Vertigan et al. (2014b), Tables 6.7, 6.11, and 6.17, pp.65–75.



Note: The MWTP curves are a weighted average across the two classes estimated by the Institute for Choice and for a price of $75 per month.
Data source: Institute for Choice; The CIE calculations.

Fig. 1. Vertigan Panel (2014) - Choice Modelling Method, MWTP for Download Speeds, 2014 Dollars.
Benefits per month per broadband household.
Source: Extract from Vertigan et al. (2014b), Chart 6.4, p.63.



Note: The MWTP curves are a weighted average across the two classes estimated by the Institute for Choice and for a price of $75 per month.
Data source: Institute for Choice; The CIE calculations.

Fig. 2. Vertigan Panel (2014) - Choice Modelling Method, MWTP for Upload Speeds, 2014 Dollars.
Benefits per month per broadband household.
Source: Extract from Vertigan et al. (2014b), Chart 6.5, p.64.

Assuming no change in the Vertigan Panel's roll-out timings or costs, the total estimated incremental benefits of the NBN are reduced by $17–20 billion in 2014 present value terms, an amount sufficient to eliminate the Vertigan Panel's estimated net economic benefits under either the MTM or FTTP roll-out strategies (see Table 3).

4.2. Secondary methods – Actual take-up and time savings methods

Given the magnitude of the Vertigan Panel's choice modelling benefits overstatement, consideration is given to whether the Vertigan Panel's two secondary estimation methods provide any corroborative evidence to support the overstatement finding. The actual take-up

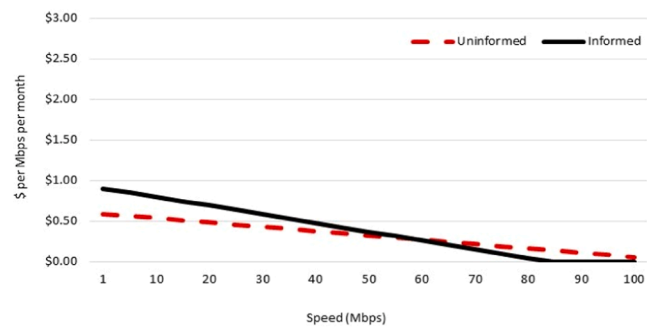


Fig. 3. Recalculated Vertigan Panel (2014) - Choice Modelling Method, MWTP for Download Speeds, 2014 Dollars.
Benefits per month per broadband household.
Source: Recalculated from Vertigan et al. (2014b), Tables H.7 and H.8, pp.178–179.

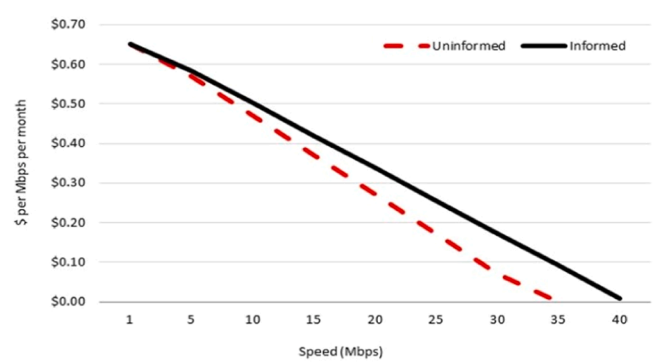


Fig 4. Recalculated Vertigan Panel (2014) - Choice Modelling Method, MWTP for Upload Speeds, 2014 Dollars.
Benefits per month per broadband household.
Source: Recalculated from Vertigan et al. (2014b), Tables H.7 and H.8, pp.178–179.

method examined the price premiums that were being paid by those households and the mix of speed plans being adopted. For the choice modelling and time savings methods, the Vertigan Panel paid close attention to the actual mix of speeds already available to households in 2014, and only estimated the benefit of incremental speed improvements. In contrast, under the actual take-up method, it was assumed that all households would benefit from access to NBN speeds of 12 Mbps, and that the benefit of speeds faster than 12 Mbps could be fully attributed to

Table 2
Vertigan Panel (2014) – Published and Recalculated Results, Choice Modelling Method, Incremental NBN Benefits per Household per Month, Fixed Line Footprint, 2014 Dollars.

Roll-out Strategy	Vertigan Panel (2014) Method: Choice Modelling Published^			Vertigan Panel (2014) Method: Choice Modelling Recalculated^^			
	DL	UL	Total	DL	UL	Total	Total % Change*
Unsubsidised Rollout	$40.40	$6.10	$46.50	$18.30	$6.10	$24.40	(48%)
MTM Rollout	$40.50	$6.10	$46.60	$18.59	$6.10	$24.69	(47%)
FTTP Rollout	$45.80	$9.70	$55.50	$20.80	$9.70	$30.50	(45%)

Benefits incremental to the "No further roll-out" strategy. DL = download speed, UL = upload speed.
^ Published results from Table 1.
^^ Recalculated results include the choice modelling error adjustment in calculating WTP for download speeds.
Source: Recreated from Vertigan et al. (2014b), Tables 6.7, 6.11, 6.17, pp.65–75, and Charts 4.4, 4.5, B.5, B.6, pp.45–46, 101. Due to need to read some Vertigan Panel data from charts, error adjusted results are indicative.

Table 3
Vertigan Panel (2014) – Published and Recalculated Results, Choice Modelling Method, NBN Total NPV Benefits, 2014 Dollars ($ billion).

Roll-out Strategy	Vertigan Panel (2014) Method: Choice Modelling Published		Vertigan Panel (2014) Method: Choice Modelling Recalculated ^^		
	Benefits NPV	Cost-Benefit NPV	Benefits NPV	Cost-Benefit NPV	Benefits Change
Unsubsidised Rollout*	$41.7	$24.0	$21.9	$4.2	($19.8)
MTM Rollout	$42.7	$17.9	$22.6	($2.2)	($20.1)
FTTP Rollout	$37.0	$1.8	$20.3	($14.9)	($16.7)

Benefits incremental to the "No further roll-out" strategy, including both fixed Line and non-fixed Line footprints. NPV = net present value in 2014, based on the period 2014 to 2040.
^ Sourced from Vertigan et al. (2014b), Tables 1 and 6.3, p.10 and p.61.
^^ Recalculated results assume total benefit reductions proportional with the per household reductions presented in Table 2, and no change in costs or rollout footprint from the Vertigan Panel's published results for each roll-out scenario.
* Note that under the unsubsidised roll-out, net benefits would likely be higher if the roll-out footprint was reduced.

the NBN. In 2014, however, based on the Vertigan Panel report, about 41 percent of households already had access to speeds of about 12 Mbps or higher.[15] After adjusting for the oversight in the Vertigan Panel's estimates, the actual take-up method benefits per household are reduced by about 48 percent (see details in Appendix B).

With the time savings method, the Vertigan Panel relied on a bottom-up model of forecast household bandwidth requirements to estimate the degraded minutes that households would experience if available speeds were not improved.[16] Each forecast degraded minute was valued using a rate of $15 per hour,[17] resulting in an average household benefit of $53 per month in the fixed-line footprint, marginally higher than the choice modelling estimate of $46 per month (see Table 1). There are several reasons to believe that the time savings method benefits were likely to be high-end estimates. First, every degraded minute was valued, whereas it is likely that some degraded minutes would either not be noticed or not inconvenience users. Second, in valuing degraded minutes at $15 per hour, a rate typically used to represent the opportunity cost of paid work, the estimates assumed the same hourly cost for all members of the population. And third, unlike the other methods, the time savings method estimates were not constrained by household income limits. This issue is particularly pertinent given that the benefits of the time-savings method were expected to highly concentrated amongst households with access to the slowest broadband speeds in 2014 (see details in Appendix C, Fig. C.1). If it is accepted that the time savings approach had characteristics that tended to generate high-end benefit estimates, then the closeness of the Vertigan Panel's time savings and choice modelling benefit estimates is surprising, and provides further support for the finding that the choice modelling benefits were overstated.

5. Updated high-speed broadband benefits in 2022

For updated insights on the benefits of high-speed broadband a simplified version of the Vertigan Panel's stated choice experiment is repeated in 2022. For brevity, where comparisons with the Vertigan Panel are presented, they focus on the MTM roll-out strategy, which is the actual strategy deployed by the NBN, and the fixed line footprint, which comprises 95 percent of NBN's connected premises in 2022. Further, comparisons with the Vertigan Panel study focus on the incremental benefits of the NBN per broadband household, rather than the total NPV benefits, as this study does not revisit the NBN costs or roll-out timings.

5.1. Choice modelling experiment design

The choice experiment for this study follows a similar design to that used by the Vertigan Panel. The survey commences with several respondent qualifier questions. This is followed by questions on the respondent's household demographics profile, broadband subscription details, and broadband usage characteristics. By asking respondents to recall their existing service and usage information, this part of the survey supports cognitive build-up, leading to more reliable responses in the later choice tasks. The respondents are then randomly split into two groups. An uninformed group receives no information on broadband speeds. An informed group receives information on the speed requirements of common broadband activities, similar in content to the information aid used by the Vertigan Panel (see Appendix D, Fig. D.1).

Each respondent is presented with eight choice tasks, with each task including four choice options (see Fig. D.2). Each option includes five attributes: access technology; data allowance; download speed; upload speed; and price (see Table 4). The attribute levels for access technology and data allowance do not vary across the choice options, with all options having "Fixed Line" as the technology and "Unlimited" as the allowance. These two attributes, although they do not vary, are included to ensure consistency with the way information is presented by service providers in the respondent market. Whilst additional attributes could be included, the included attributes closely approximate consumer choices in 2022. To mitigate any risks associated with attribute exclusion, the survey takes a similar approach to Liu et al. (2018), stating that the respondents should assume that any omitted features or inclusions

[15] Vertigan Panel (2014b), Chart 4.4, p.45, and Chart B.5, p.101.
[16] Note that at the time of the Vertigan Panel report, the forecast bandwidth estimates were publicly criticized for being too conservative. A brief discussion of this criticism is included in Appendix C.
[17] Vertigan Panel (2014b), p.67.

Table 4
This Study (2022) – Choice Experiment Attributes, Descriptions and Levels.

Attribute	Description	No. of Levels	Level Descriptions
Connection	The type of broadband access technology.	1	Fixed Line
Data Allowance	The monthly data allowance cap.	1	Unlimited
Download Speed	The maximum download speed for the service.	8	5 Mbps, 12 Mbps, 25 Mbps, 50 Mbps, 100 Mbps, 150 Mbps, 250 Mbps, 500 Mbps
Upload Speed	The maximum upload speed for the service.	5	1 Mbps, 5 Mbps, 10 Mbps, 20 Mbps, 40 Mbps
Price (monthly)	The monthly household subscription price.	12	$40-$150 - $10 increments

are identical to the respondent's existing home broadband service.

As recommended by prior researchers, hypothetical bias risk is minimised by designing the choice tasks to be as realistic as possible (Liu et al., 2018; Ben-Akiva et al., 2019). First, the options are designed to resemble the choices presented to consumers in the market. Second, the respondents are requested to select a preferred option as if purchasing for their own household. And third, after making their initial choice, the respondents are asked to choose between their selected choice and their actual home broadband subscription.

5.2. Data collection and descriptive statistics

The survey respondents are sourced in May and June 2022 from Qualtrics, a global survey administrator that sources respondent samples from multiple panel providers. To ensure data quality, respondents are excluded from completing the survey if they are less than 18 years old, not the primary decision-maker for home broadband services, or from a household that does not subscribe to home broadband (34 percent disqualified). Further, respondents are excluded where they are found to be straight-line respondents or complete the survey in less than half the median completion time (additional 9 percent disqualified). After applying the above quality controls, the final data set includes 668 responses with a median completion time of 14.8 min. As each respondent completes eight choice tasks, there are 5344 completed tasks. The respondents receive a monetary reward for completing the survey.

The sample respondents are selected to be representative of the Australian population. As with most households, the respondents primarily subscribe to fixed line broadband, with speeds of 25, 50 or 100 Mbps and unlimited data allowances (see Appendix E, Table E.2). As the respondents are required to be the primary decision-makers for home broadband services, however, they tend to be slightly older and from larger households (see Table E.1). For both the demographic and broadband subscription profiles, the respondent mixes are similar across the informed and uninformed groups. In addition, when presented with the status quo question, 45–46 percent of the respondents in each group select their preferred hypothetical choice option, indicating that the hypothetical options provide realistic alternatives (see Table E.3).

5.3. Econometric method and results

The survey respondent stated choices are analysed with a random utility model (McFadden, 1973), where respondents are assumed to maximize household utility. Taking a similar approach to Liu et al. (2018), each download and upload speed interval is treated as a separate dummy variable in the model, removing any restriction on the change in utility being linear between speed intervals. A conditional logit model is used with the following specification:

$$U = \mathrm{B}p\mathrm{x}Price + \mathrm{B}_1\mathrm{x}DL12Mbps + \mathrm{B}_2\mathrm{x}DL25Mbps + \mathrm{B}_3\mathrm{x}DL50Mbps + \mathrm{B}_4\mathrm{x}DL100Mbps + \mathrm{B}_5\mathrm{x}150Mbps + \mathrm{B}_6\mathrm{x}DL250Mbps + \mathrm{B}_7\mathrm{x}DL500Mbps + + \mathrm{B}_9\mathrm{x}UL5Mbps + \mathrm{B}_{10}\mathrm{x}UL10Mbps + \mathrm{B}_{11}\mathrm{x}UL20Mbps + \mathrm{B}_{12}\mathrm{x}UL40Mbps + e \quad (1)$$

U = utility of choice option
$\mathrm{B_p}$ = utility parameter for price
$\mathrm{B_1}$-$\mathrm{B_{13}}$ = utility parameter for download and upload speed intervals
DL = download speed interval (measured relative to base speed of 5 Mbps)
UL = upload speed interval (measured relative to base speed of 1 Mbps)
e = random error

The WTP for each download and upload speed is calculated by dividing the utility parameter for each broadband speed alternative by the price parameter:

$$\mathrm{WTP} = -\mathrm{B}_{1-13}/\mathrm{B_p} \quad (2)$$

WTP = willingness to pay
$\mathrm{B_1}$-$\mathrm{B_{13}}$ = utility parameter for download and upload speed intervals
$\mathrm{B_p}$ = utility parameter for price

Consistent with prior research, households are willing to pay more for faster broadband speeds, but incrementally less for each Mbps increase (see Table 5). Using average results across the informed and uninformed groups, households report a MWTP of $1.45 per Mbps to increase download speeds from 5 to 12 Mbps ($10 total), $1.65 per Mbps for 12 to 25 Mbps ($21), $0.94 per Mbps for 25 to 50 Mbps ($24), $0.17 per Mbps for 50 to 100 Mbps ($9), -$0.04 per Mbps for 100 to 150 Mbps (-$2), $0.05 per Mbps for 150 to 250 Mbps ($5), and $0.03 per Mbps for 250 to 500 Mbps ($7). These findings are broadly consistent with those reported by other studies that have used stated preference techniques. Liu et al. (2018) found that households had a MWTP of US$2.34 per Mbps at download speeds of 4 to 10 Mbps (this study $1.45), declining to US$0.02 per Mbps for speeds of 100 Mbps to 1 Gbps (this study $0.025 from 100 to 500 Mbps). And Worden and Hambly (2022) found that households were WTP about an extra $10 per month to improve broadband download speeds from 50 Mbps to 100 Mbps (this study $8.66) and about an extra $15 to increase speeds from 100 Mbps to 500 Mbps (this study $10.65).[18] The results also show consistency with studies using revealed preference approaches. Nevo et al. (2016), examining household demand at a time when average download speeds were 14.7 Mbps, found an average WTP of $2.03 per Mbps increase in speed (this study $1.45 between 5 and 12 Mbps). And Mendez et al. (2021), at a time when most households had download speeds of 25 Mbps or higher, found high-income (low-income) households willing to pay $0.93 ($0.60) per Mbps increase in speed (this study $0.94 between 25 and 50 Mbps).

An anomaly in this study is the slight decline in the WTP for speed increases from 100 to 150 Mbps, which may be due to a tendency for service providers to promote the 100 Mbps plan, possibly combined with a consumer preference for rounded numbers.[19] Further, although respondents in the informed group report a higher MWTP for download speeds (relative to the uninformed group), this is offset by a lower MWTP for upload speeds. Whilst this finding is assumed to be due to the informed group information aid emphasising the relative importance of download speeds, a similar offset between download and upload speeds was not identified by the Vertigan Panel.

[18] See Worden et al. (2022), Table 4.
[19] Note that Peterson (2017) identified a similar anomaly in analysing WTP changes between speed levels.

Table 5
This Study (2022) - Choice Modelling Method, Parameter Estimation Results, 2022 Dollars.

	Uninformed Group			Informed Group			Average WTP^	
Variables	Coefficient	Total WTP	s.e.	Coefficient	Total WTP	s.e.	Total	per Mbps
Price	−0.0355			−0.0355				
Download Speed								
12 Mbps	0.3148	$8.88	$5.48	0.4055**	$11.42**	$5.16	$10.15	$1.45
25 Mbps	0.9229***	$26.02***	$6.79	1.3214***	$37.22***	$6.55	$31.62	$1.65
50 Mbps	1.7214***	$48.54***	$7.51	2.1967***	$61.87***	$7.35	$55.20	$0.94
100 Mbps	2.0374***	$57.45***	$7.80	2.4952***	$70.28***	$7.63	$63.86	$0.17
150 Mbps	1.9488***	$54.95***	$8.30	2.4488***	$68.97***	$8.19	$61.96	($0.04)
250 Mbps	2.1486***	$60.59***	$8.52	2.6305***	$74.09***	$8.40	$67.34	$0.05
500 Mbps	2.3360***	$65.87***	$8.81	2.9523***	$83.15***	$8.71	$74.51	$0.03
Upload Speed								
5 Mbps	0.4360***	$12.29***	$4.47	0.1365	$3.84	$4.37	$8.07	$2.02
10 Mbps	0.8058***	$22.72***	$5.65	0.2677	$7.54	$5.55	$15.13	$1.41
20 Mbps	1.1065***	$31.20***	$6.05	0.4977**	$14.02**	$5.97	$22.61	$0.75
40 Mbps	1.3318***	$37.55***	$6.16	0.8039***	$22.64***	$6.06	$30.10	$0.37

WTP measured per month per broadband household, download speeds relative to 5 Mbps and upload speeds WTP relative to 1 Mbps.
^ Average calculated across informed and uninformed groups. Per Mbps calculated as the incremental WTP per Mbps to move from the speed tier immediately below the relevant speed tier.
*$p<0.1$.
** $p<0.05$.
*** $p<0.01$.

Comparing the results of this study (in 2014 dollars) with the recalculated Vertigan Panel choice modelling results from 2014, there has been an increase in the WTP for both download and upload speeds. Across the informed and uninformed groups, the real compound annual growth rate (CAGR) in household WTP has been 8.8 percent for download speeds and 11.5 percent for upload speeds, with reasonably consistent growth across all speed tiers (see Table 6, Figs. 5 and 6). Note that whist the comparisons in this section focus on the recalculated Vertigan Panel choice modelling results, Fig. 5 includes both the

Table 6
Vertigan Panel (2014) vs. This Study (2022) – Choice Modelling Method, Total WTP per Household, 2014 Dollars.

	Average Total WTP^		Real Growth		
Variables	Vertigan Panel	This Study	Increase	Total%	CAGR %
Download Speed					
12 Mbps	$4.59	$8.58	$3.98	86.7%	8.1%
25 Mbps	$12.13	$26.73	$14.60	120.4%	10.4%
50 Mbps	$22.99	$46.67	$23.68	103.0%	9.3%
100 Mbps	$30.54	$53.99	$23.44	76.7%	7.4%
150 Mbps	*$30.54*	$52.38	$21.83	71.5%	7.0%
250 Mbps	*$30.54*	$56.92	$26.38	86.4%	8.1%
500 Mbps	*$30.54*	$62.99	$32.44	106.2%	9.5%
Average*	–	–	–	96.7%	8.8%
Upload Speed					
5 Mbps	$2.45	$6.82	$4.37	177.9%	13.6%
10 Mbps	$5.11	$12.79	$7.68	150.1%	12.1%
20 Mbps	$9.07	$19.11	$10.04	110.8%	9.8%
40 Mbps	$11.52	$25.44	$13.92	120.8%	10.4%
Average*	–	–	–	139.9%	11.5%

WTP measured relative to download speed of 5 Mbps and upload speed of 1 Mbps.
Vertigan Panel results created using parameter values at Vertigan et al. (2014b), Tables H.7 and H.8, pp.178–179.
Results from this study converted to 2014 dollars using inflation indexation adjustment of 1.18 for the period 2014 to 2022.
^ Average calculated across informed and uninformed groups.
* Average download and upload speed WTP growth is calculated as a straight-line average. The average for download speeds excludes the 150, 250, and 500 Mbps speed tiers, which were not part of the Vertigan Panel choice experiment. They are included in the table, assuming the Vertigan Panel WTP for the 150, 250 and 500 Mbps speed tiers is the same as for the 100 Mbps speed tier, as the Vertigan Panel reported no MWTP for download speeds faster than 100 Mbps.

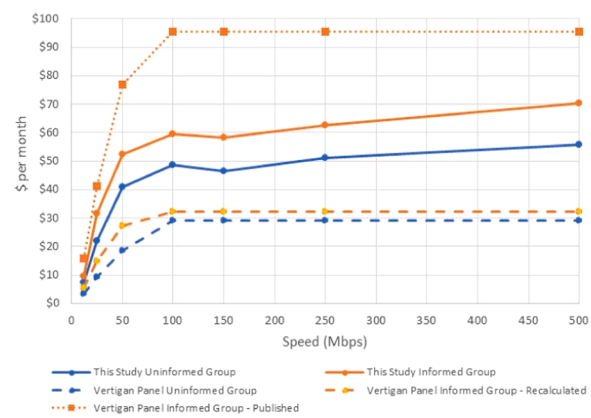


Fig. 5. Vertigan Panel (2014) vs. This Study (2022) – Choice Modelling Method, Total WTP per Household, Download Speeds, 2014 Dollars.
WTP measured relative to download speed of 5 Mbps.
Vertigan Panel results created using parameter values at Vertigan et al. (2014b), Tables H.7 and H.8, pp.178–179.
Results from this study converted to 2014 dollars using inflation indexation adjustment of 1.18 from 2014 to 2022.
Note the Vertigan Panel choice experiment did not include speed tiers 150, 250 or 500 Mbps, which are included in the figure assuming the Vertigan Panel WTP for these speed tiers is the same as for the 100 Mbps speed tier, as the Vertigan Panel reported no MWTP for speeds faster than 100 Mbps.

Vertigan Panel's published and recalculated download WTP results for informed respondents. Underscoring the magnitude of the error in the Vertigan Panel report, the corresponding WTP results from this study in 2022 remain substantially lower than the Vertigan Panel's published results from 2014.

In calculating the incremental benefits of the NBN in 2022, the NBN's actual broadband subscription speed mix is used.[20] This mix reflects the actual speeds being experienced by households, thereby recognising that some households are subscribing to speeds slower than the maximum attainable speeds available at their premise, as well as allowing for the fact that a small proportion of households are subscribing to speeds faster than 100 Mbps (which was the maximum available speed assumed

[20] Source: ACCC (2022).

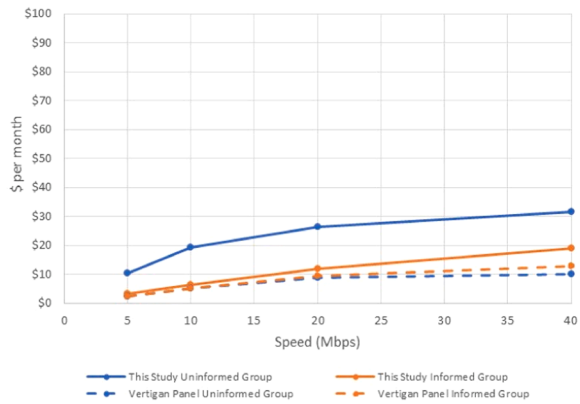


Fig. 6. Vertigan Panel (2014) vs. This Study (2022) – Choice Modelling Method, Total WTP per Household, Upload Speeds, 2014 Dollars.
WTP measured relative to upload speed of 1 Mbps.
Vertigan Panel results created using parameter values at Vertigan et al. (2014b), Tables H.7 and H.8, pp.178–179.
Results from this study converted to 2014 dollars using inflation indexation adjustment of 1.18 for the period 2014 to 2022.

by the Vertigan Panel). The analysis finds that the incremental broadband benefits enabled by the NBN are $37.40 per month per household in 2022 (in 2014 dollars), a real CAGR of 5.3 percent from the recalculated Vertigan Panel estimate from 2014 (see Table 7). Despite this growth, however, the benefits per household remain materially lower than those published by the Vertigan Panel in 2014.

Some differences in approach between this study and the Vertigan Panel study should be noted. First, this study estimates the WTP for broadband speeds using ordinal variables (modelled with multiple dummy variables), whereas the Vertigan Panel estimated the WTP using continuous variables. Whilst this difference may impact the comparability of the Vertigan Panel's estimates at each broadband speed level, the difference is not expected to materially impact the average WTP calculated across the range of speed intervals. And second, this study only varies three attributes, whereas the Vertigan Panel study varied eleven attributes. Seven of the Vertigan Panel's attributes have been removed because their market share influence was not material in the Vertigan Panel's findings and the attributes are likely to have further declined in market relevance since the Vertigan Panel (see details in Appendix F, Table F.1). An exception relates to the data allowance attribute, which the Vertigan Panel identified as one of the most important attributes, having the potential to impact market share by 31–32 percent.[21] Indeed, using the Vertigan Panel's reported attribute values it can be calculated that households were willing to pay an extra $29.81 per month to move from an allowance of 100 GB per month to 1000 GB per month in 2014.[22] The attribute, however, has been held constant in this study, as all identified broadband plans were bundled with unlimited data in 2022. As households subscribing to faster plans can be expected to have higher data transfer demands, it can be presumed that the increase in WTP for broadband speeds since the Vertigan Panel is in part due to plans becoming bundled with unlimited data between 2014 and 2022.

6. Heterogeneity of high-speed broadband benefits in 2022

For additional insight, consideration is given to the variation in WTP for broadband by household type. To enable comparability of results, this study takes a similar approach to Liu et al. (2018) and analyses common demographic variables by calculating the WTP difference between opposing survey respondent groupings (see Table 8). As found by Liu et al. (2018), higher income households are willing to pay a premium for faster download speeds. In contrast to Liu et al. (2018), however, households are also willing to pay a premium if they have children ($43.23 for 100 Mbps in this study, compared to a discount of $11.65 in Liu et al., 2018), are younger ($17.79, compared to a discount of $3.27), or in urban locations ($17.23, compared to a discount of $7.22). Although not reported by Liu et al. (2018), a WTP premium is also identified in this study where the household is large ($40.01 per month for 100 Mbps) or the primary broadband decision-maker is male ($6.57).

Whilst the above results confirm the heterogeneity in the WTP for high-speed broadband, the change in WTP premiums between this study and Liu et al. (2018) is striking. Whilst it is beyond the scope of this study to discern the precise reasons for the change, two possible explanations are offered. The first is that internet usage patterns have changed over the 5 years from 2017 to 2022, possibly induced in part by the impact of COVID-19 and increased working from home, whereby households that are younger, with children, and urban, are now prepared to pay a premium for faster speeds. The second is that the changes in the structure of broadband plans have resulted in households altering their relative WTP for broadband speeds. More specifically, the bundling of unlimited data in broadband plans may have resulted in some households altering the premium they are willing to pay for faster speeds.

To further understand the variation in WTP for high-speed broadband, the results are diagnosed by household size and income. The analysis reveals that both single-occupant and low-income households report negligible WTP for speeds faster than 50 Mbps, and that two-occupant and medium-income households report little interest in speeds above 100 Mbps (see Table 9). Only where households have three or more occupants, or are high-income households, is there a material interest in paying extra for speeds faster than 100 Mbps. Whilst these findings are based on broadband demand in 2022, and might therefore be considered lower-bound WTP estimates, the high heterogeneity in WTP is expected to create challenges for policymakers and broadband providers seeking to optimize the value of infrastructure investments. The results provide support for Lindlacher's (2021) finding that higher-speed broadband offerings are only likely to be of interest to specific household segments. Further, they indicate that maximizing the take-up of faster speeds by households with a lower WTP is likely to be at least partially dependant on the ability of broadband service providers to deploy effective price discrimination strategies.

7. Results discussion

7.1. The WTP for high-speed broadband

The build of the NBN has been one of the most significant public infrastructure projects undertaken in Australia. The Vertigan Panel was established to perform a cost-benefit analysis of alternative NBN roll-out strategies. Composed of an eminent group of experts and supported by multiple research teams, the Vertigan Panel was well equipped to perform the task. At 200 pages in length, and including three benefit estimation methods, the Vertigan Panel delivered a report that presented as comprehensive and thorough. This study has found, however, that the Vertigan Panel's economic benefit estimates were materially overstated.

An analysis of the Vertigan Panel's primary benefit estimates, which were supported by choice modelling, finds that they were overstated by 47 percent, or $17–20 billion in 2014 present value terms (see Table 3), an amount sufficient to eliminate the estimated net economic benefits from the NBN. Further, a review of the Vertigan Panel's secondary methods indicates that they had characteristics that are likely to have

[21] Range based on total attribute importance for uninformed (31 percent) and informed (32 percent) respondent groups. Source: Vertigan Panel (2014b), Table H.9, p.181.

[22] Calculated using Vertigan Panel methodology and attribute values. Source: Vertigan Panel (2014b), p.176-178.

Table 7
Vertigan Panel (2014) vs. This Study (2022) – Choice Modelling Method, Incremental NBN Benefits per Household per Month, Fixed Line Footprint, 2014 Dollars.

Roll-out Strategy	Vertigan Panel Publish Results 2014 and 2022			Vertigan Panel Recalculated 2014 and This Study 2022		
	2014 Published^	2022 "Published"^	Real CAGR	2014 Recalculated^	2022 Actual^^	Real CAGR
MTM Rollout	$46.60	$54.60	2.0%	$24.69	$37.40	5.3%

^ Benefits based on the difference between the attainable speeds in 2014 and forecast attainable speeds under the MTM roll-out. Vertigan Panel published 2014 and recalculated 2014 results from Table 2. Vertigan Panel "published" results for 2022 created using the Vertigan Panel's real growth rate assumption of 2 percent per annum, source: Vertigan et al. (2014b), p.77.

^^ Benefits based on the difference between the attainable speeds in 2014 and the NBN subscribed speeds in June 2022, converted to 2014 dollars using inflation indexation adjustment of 1.18 for the period 2014 to 2022. The analysis assumes that the benefits of the attainable speeds available in 2014 have increased at the WTP growth rates reported in Table 6.

Table 8
This Study (2022) - Choice Modelling Method, WTP Difference by Demographic Group, per Household per Month, 2022 Dollars.

Variables	Urban	Young	High Income	Children	Large	Gender Male
Download Speed						
12 Mbps	-$0.79	-$1.61	$3.40*	-$2.73	$11.50*	-$3.76
25 Mbps	$1.69***	$9.28***	$8.15***	$17.06***	$16.99***	-$0.22***
50 Mbps	$8.30***	$16.12***	$19.81***	$30.18***	$31.11***	$6.13***
100 Mbps	$17.23***	$17.79***	$33.54***	$43.23***	$40.01***	$6.57***
150 Mbps	$20.38***	$26.56***	$34.45***	$48.21***	$40.10***	$4.88***
250 Mbps	$19.85***	$29.81***	$42.20***	$50.84***	$43.23***	$4.21***
500 Mbps	$25.24***	$36.17***	$45.24***	$54.34***	$47.73***	$9.23***
Upload Speed						
45 Mbps	$1.27	-$0.22	$2.99*	$5.62	-$0.54	-$0.79
10 Mbps	-$7.94**	-$7.73	$1.66**	-$7.55	-$5.27	-$1.21**
20 Mbps	-$14.58***	-$6.38**	$1.61***	-$5.69	-$3.65*	-$0.36***
40 Mbps	-$14.22***	-$5.07***	-$0.03***	-$0.01**	$1.78***	-$1.30***
% of Respondents	80%	50%	50%	32%	30%	48%

Urban includes households in major cities, identified using respondent postcode. Young includes decision-makers less than 47 years old (survey respondent median). High income includes households with income more than $1625 per week (survey respondent median). Children includes all households with children. Large includes households with 4 or more occupants (survey median was 3 people). Download relative to 5 Mbps, Upload relative to 1 Mbps.

** $p<0.05$.
*** $p<0.01$.
* $p<0.1$.

Table 9
This Study (2022) - Choice Modelling Method, WTP by Demographic Group, per Household per Month, 2022 Dollars.

	Household Size				Household Income		
Variable	One	Two	Three	Four or More	Low	Med	High
Download Speed							
12 Mbps	$7.48	$16.65***	$0.07	$20.08*	$6.61	$15.94**	$13.02
25 Mbps	$23.83***	$36.75***	$28.48**	$45.85***	$23.70***	$37.29***	$41.46***
50 Mbps	$41.04***	$57.44***	$55.14***	$80.59***	$40.57***	$63.80***	$70.20***
100 Mbps	$37.15***	$63.69***	$75.97***	$96.09***	$40.98***	$74.21***	$85.40***
150 Mbps	$33.23***	$62.61***	$73.86***	$94.22***	$38.92***	$71.45***	$84.52***
250 Mbps	$35.64***	$67.51***	$78.18***	$101.68***	$44.30***	$69.17***	$95.07***
500 Mbps	$43.10***	$68.77***	$91.14***	$112.22***	$45.27***	$79.61***	$104.72***
Upload Speed							
5 Mbps	$8.66*	$6.54	$6.94	$7.44	$7.63*	$7.28	$8.25
10 Mbps	$8.60	$21.89***	$9.24	$10.60	$11.24**	$17.50**	$16.21*
20 Mbps	$16.80**	$27.82***	$15.36	$19.13*	$20.82***	$21.62***	$24.09***
40 Mbps	$22.15***	$34.71***	$21.24*	$30.65***	$26.82***	$31.12***	$31.75***
% of Respondents	18%	32%	20%	30%	29%	32%	40%

Household income low = <$1000 per week, medium = $1000–1999 per week, high = $2000 or more per week. Download relative to 5 Mbps, Upload relative to 1 Mbps.

** $p<0.05$.
*** $p<0.01$.
* $p<0.1$.

produced high-end benefit estimates, thereby providing corroborative evidence that the choice modelling benefits were overstated. Collectively, the findings provide evidence that the government's decision to invest in the NBN in 2014 was based on erroneous information.

The choice modelling analysis in this study finds that the WTP per household has experienced a real CAGR of 8.8 percent for download speeds and 11.5 percent for upload speeds between 2014 and 2022 (Table 6). These rates of growth in the WTP for broadband, which are the first to be reported since the completion of the transition from dial-up to broadband, are around the lower-end of the implied 10–17 percent annual rates identified in earlier studies by Dutz et al. (2009) and Rosston et al. (2010). Despite the above growth rates, however, the actual incremental benefits of the NBN per household, calculated using actual subscribed speeds in 2022, remain materially lower than those predicted

by the Vertigan Panel in 2014.

Caution should be exercised in interpreting the above WTP growth rates. This is because they have been assisted by changes in broadband plan offerings since 2014, whereby plans are now bundled with unlimited data. As an example, whilst the WTP for broadband plans with download speeds of 50 and 100 Mbps each increased by about $23 per month between 2014 and 2022 (in 2014 dollars, see Table 6), these download speeds come bundled with unlimited data in 2022, an attribute value that households were willing to pay at least an extra $30 per month for in 2014.[23] This change in broadband offerings, which was not anticipated in the Vertigan Panel report, will have supported the growth in the WTP for broadband speeds, as faster speeds are more valuable to households when coupled with unlimited data. Without comparable future changes in plan constructs, it is reasonable to expect that the rates of growth in the WTP for broadband speeds may decline from those reported in this study.

The results of this study are consistent with earlier studies, finding that households are willing to pay more for faster speeds, but incrementally less for each additional Mbps. Providing up-dated insights since Liu et al. (2018), the results provide evidence that the WTP for broadband is highest for households that have higher incomes, are larger, have children, are younger, and are urban. Further, consistent with the insights of Lindlacher (2021), the results suggest that some household segments do not place any material value on download speeds much above 50 Mbps, and that the take-up of higher speeds may be limited to larger, higher-income households. And as foreshadowed by Ergas and Robson (2009), and recently explored by Mendez et al. (2021) in the context of broadband adoption, the high WTP heterogeneity identified in this study indicates a potential future role for price discrimination in optimizing the value of broadband investments.

7.2. Cost-benefit analysis impacts and lessons

It is worth considering the significant impacts of the Vertigan Panel findings. The benefit estimates provided strong economic justification for the government's investment in the NBN, as well as an endorsement of the MTM deployment strategy. The estimates also inadvertently provided the government with justification for NBN Co continuing to operate as a monopoly, government-owned business entity. As argued by the responsible government Minister at the time, modifying the NBN's then structure or regulatory arrangements to increase competition, as had been recommended by the Vertigan Panel, would have distracted NBN's management and risked slowing the delivery of the NBN's benefits.[24] Further, it is not unreasonable to contend that the strength of the Vertigan Panel's findings, and their role in shaping media and public opinion, have assisted in providing ongoing legitimacy over subsequent years to the government's NBN investment and regulatory decisions.

This study has no evidence that can be used to determine whether the Vertigan Panel's estimation errors were in any way influenced by intentional misrepresentation or manipulation, by either the Vertigan Panel or any of its agents. Putting aside this possibility, however, the Vertigan Panel does appear to have overlooked data inconsistencies that may have served as warnings that the benefits were overstated. For example, it did not examine the substantial difference in WTP for download speeds between informed and uninformed respondents, or scrutinise the proximity of the benefit estimates generated using alternative estimation techniques, and it appears to have paid little attention to then available international research indicating a rapidly declining MWTP for faster speeds (see details in Appendix G). It is also noteworthy that the Vertigan Panel's findings were not subject to any formal peer review when released and have not been subject to any review since.

It is worth reflecting on whether the primary benefit estimation technique used by the Vertigan panel was robust. This is of interest, as the Vertigan Panel's choice analysis was conducted at a time when household experience with high-speed broadband was limited, and many of the services that would come to depend on high-speed broadband for delivery were yet to become widely available. Despite this context, however, the choice technique appears to have provided the Vertigan Panel with valuable evidence of the declining MWTP for high-speed broadband, which was pivotal in assessing the relative benefits of the MTM and FTTP roll-out strategies. Further, if it were not for the Vertigan Panel's calculation error, the technique would have alerted decision-makers to the risk that the net economic benefits of the NBN were likely to be negligible. It is also noteworthy that the choice modelling technique could have been used to provide insights on the heterogeneity of demand, which may have served as a further warning on the challenges that the NBN, as an entity restricted to providing uniform national pricing, would likely face in maximizing the household take-up of faster speeds. On the other hand, there is no indication that the technique could have predicted changes in the relative importance of different broadband attributes, which has likely been a factor contributing to the high rate of growth in the WTP for speeds since 2014. Apart from this limitation, however, choice modelling had much to offer decision-makers in 2014, even though high-speed broadband was then an emerging product with changing characteristics and uncertain demand.

Although cost-benefit analyses provide a logical way to evaluable public infrastructure investments, this study demonstrates that such analyses, no matter how well resourced and sophisticated, can be prone to error. It is worth considering the potential lessons that might be drawn from the Vertigan Panel experience. First, although the Vertigan Panel released granular information, the information was not sufficiently detailed or transparent for a cost-benefit analysis practitioner to easily trace and repeat. Whereas Bellas and Kosnik (2019) has recommended that practitioners keep replicable records of all numerical calculations, the Vertigan Panel experience suggests that practitioners should go one step further and make available their analyses in detailed and accessible formats, to enable thorough, efficient, and timely scrutiny. Second, whilst the Vertigan Panel encouraged future scrutiny of its findings, it did not formally recommend any mechanisms or timeframes for such activities. Whereas Flyvbjerg and Bester (2021) has suggested cost-benefit analyses should be audited, the Vertigan Panel experience suggests that practitioners should be specific about future audits or reviews, in terms of the timings and approach, to improve the probability that such activities will take place. And third, although the Vertigan Panel recognised the risks associated with a substantial investment of taxpayer funds, it did not highlight the benefit risks associated with calculation error or unconscious bias. Where cost-benefit analyses are being used to support extensions of government ownership and control, thereby displacing the corrective role of competitive markets, alerting users of these risks would appear to be highly pertinent for future practitioners.

There are several limitations to this study. As discussed earlier, there are differences in the choice modelling method applied by the Vertigan Panel and this study. Whilst some of these differences are due to changes in the broadband market, others have been driven by a desire to either simplify the choice experiment or apply more contemporary analysis techniques. Further, whilst reporting robust growth in the WTP for broadband speeds between 2014 and 2022, it is not possible to determine whether the growth has been at a consistent linear rate or uneven, and nor is it possible to discern the precise drivers of the growth. In addition, in assessing the benefits of the NBN, the analysis focuses on

[23] Estimate based on the Vertigan Panel WTP results for an included data allowance increase from 100 GB to 1000 GB per month.

[24] Source: Communications Minister Malcolm Turnbull, reported in Ramli (2014).

private household benefits, and does not revisit the Vertigan Panel's finding that the incremental business, public, and external benefits of high-speed broadband were likely to be low.[25] These limitations, most of which are not easily avoided, demonstrate the challenges with performing ex-post cost-benefit analysis comparisons. Despite these limitations, however, the findings provide decision-makers with a contemporary up-date on the WTP for high-speed broadband, as well as insights on the adoption of WTP techniques in a policy setting.

8. Conclusion

As many governments embark on ambitious plans to improve household access to faster broadband speeds, the findings of this paper may serve as a cautionary tale. Although able to inform and enhance decision-making, cost-benefit analyses also carry the risk of estimation error. In the case of the NBN, it is unfortunate that a single error, buried deep in complex analysis, may have been instrumental in offering the economic justification for one of Australia's largest public infrastructure projects. Whilst more ex-post cost-benefit analysis reviews are required, the study findings suggest that practitioners need to be vigilant in ensuring their results are robust and capable of being efficiently scrutinised. Further, given the risk of error, policy decision-makers should exercise caution where cost-benefit analyses are being relied upon to justify extensions of government that enduringly replace the dynamics of market decision-making. Indeed, competitive markets, where they can be acquiesced, may provide the best defence against cost-benefit estimation errors, and be far superior to the promise of future audits and reviews.

Whilst the study findings indicate a healthy increase in the WTP for broadband speeds over the period 2014 to 2022, headwinds for broadband infrastructure providers have been identified. Analysis indicates that recent growth in the WTP for broadband speeds may have been supported by changes to broadband plan offerings, and that without similar future changes the growth in WTP may slow. Further, there is evidence of high heterogeneity in the WTP for faster broadband, with many households unwilling to pay a premium for speeds much faster than about 50 Mbps. When coupled with the risk of cost-benefit analysis error, these findings indicate that the future role for governments in funding broadband infrastructure investments and regulating broadband pricing should be approached with increased caution. Moreover, the NBN experience suggests that government broadband investment models that seek to wholly displace the role of markets in guiding infrastructure roll-outs carry a heightened risk of exposing taxpayers to significant losses.

CRediT authorship contribution statement

Andrew Sinclair: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

Appendix A. Recalculation of Vertigan Panel's Choice Modelling Method Results

Figs. A.1 and A.2 present the choice experiment attribute values reported by the Vertigan Panel. The four sets of values inside the bold boxes were used by the Vertigan panel to calculate the MWTP for download and upload speeds. Included are the attribute values for the two respondent groups, each with two latent classes. (Note that Figs. A.1 and A.2 also include status quo attribute values, which were not used by the Vertigan Panel in determining the MWTP estimates.)

Attribute	Class 1				Class 2			
	Status Quo		New Plans		Status Quo		New Plans	
	parameter	*t-ratio*	*parameter*	*t-ratio*	*parameter*	*t-ratio*	*parameter*	*t-ratio*
Contract	0.1282	1.88	0.0000	N/S	0.1103	1.71	0.0737	2.01
Data	0.1927	5.74	0.3773	10.06	0.1165	3.66	0.3646	15.06
Restrictions	-0.1704	2.20	0.0000	N/S	-0.1589	2.10	-0.0937	2.79
Price	-0.0092	3.17	-0.0599	7.41	-0.0086	4.46	-0.0073	1.70
Price (Sqr)	0.0000	N/S	0.0001	3.10	0.0000	N/S	-8.88E-05	4.32
Download	0.0175	2.57	0.0050	3.15	0.0231	7.37	0.0221	3.87
Download (Sqr)	0.0000	N/S	0.0000	N/S	0.0000	N/S	-1.10E-04	2.49
Upload	0.0575	2.89	0.0000	N/S	0.0000	N/S	0.0289	3.28
Upload (Sqr)	0.0000	N/S	0.0000	N/S	0.0000	N/S	-4.30E-04	2.08
Home phone	0.0000	N/S	0.2073	3.55	0.3167	4.22	0.2220	6.42
Pay TV	0.2647	1.91	-0.1067	2.03	0.2563	2.54	0.0565	1.91
Mobile	0.3960	3.91	0.0000	N/S	0.1588	1.87	0.0000	N/S
WiFi Modem	0.0000	N/S	0.0000	N/S	0.0000	N/S	0.0000	N/S

Attribute values relevant to recreating Vertigan Panel MWTP figures in bolded boxes.
Source: Extract from Vertigan Panel (2014b), Table H.7, p.178.

Fig A.1. Vertigan Panel (2014) - Choice Modelling Method, Attribute Values for Uninformed Survey Respondents.
Attribute values relevant to recreating Vertigan Panel MWTP figures in bolded boxes.
Source: Extract from Vertigan et al. (2014b), Table H.7, p.178.

Attribute	Class 1				Class 2			
	Status Quo		New Plans		Status Quo		New Plans	
	parameter	*t-ratio*	*parameter*	*t-ratio*	*parameter*	*t-ratio*	*parameter*	*t-ratio*
Contract	0.0000	N/S	0.0000	N/S	0.0000	N/S	0.0724	1.89
Data	0.1431	3.74	0.5549	12.27	0.0784	2.53	0.2751	10.99
Restrictions	-0.1625	1.96	0.0000	N/S	-0.2567	3.25	-0.0683	2.03
Price	-0.0302	5.26	-0.0712	8.37	-0.0224	5.82	-0.0226	5.16
Price (Sqr)	6.84E-05	3.65	2.10E-04	5.36	4.28E-05	3.47	0.0000	N/S
Download	0.0000	N/S	0.0274	2.47	0.0601	5.95	0.0255	4.22
Download (Sqr)	0.0000	N/S	-2.00E-04	2.25	-3.50E-04	3.35	-1.30E-04	2.72
Upload	0.0000	N/S	0.0000	N/S	0.0000	N/S	0.0357	3.88
Upload (Sqr)	0.0000	N/S	0.0000	N/S	0.0000	N/S	-4.40E-04	2.05
Home phone	0.2523	3.17	0.1380	2.06	0.2029	3.10	0.2080	5.95
Pay TV	0.3753	2.41	0.0000	N/S	0.0000	N/S	0.0611	1.95
Mobile	0.1880	1.86	0.1890	2.83	0.0000	N/S	0.0000	N/S
WiFi Modem	0.1994	2.55	0.0000	N/S	0.1586	2.58	0.0601	1.75

Attribute values relevant to recreating Vertigan Panel MWTP figures in bolded boxes.
Source: Extract from Vertigan Panel (2014b), Table H.8, p.179.

[25] See Vertigan Panel (2014a), pp.77-81.

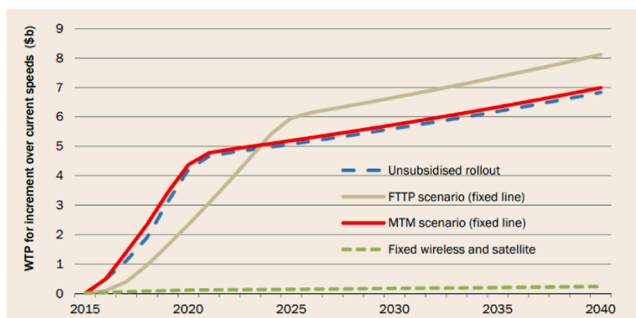
Fig A.2. Vertigan Panel (2014) - Choice Modelling Method, Attribute Values for Informed Survey Respondents.
Attribute values relevant to recreating Vertigan Panel MWTP figures in bolded boxes.
Source: Extract from Vertigan et al. (2014b), Table H.8, p.179.

Below is the formula used by the Vertigan Panel to determine MWTP. Using the attribute information in Figs. A.1 and A.2 with the formula below, Figs. 3 and 4 were created to present the recalculated MWTP results.

Marginal WTP for Broadband Speed[26] $= \frac{\beta_{sp}+2\times\beta_{sp\wedge 2}\times Sp}{\beta_{pr}+2\times\beta_{pr\wedge 2}\times \mathrm{Pr}}$

Sp = Speed in Mbps
Pr= Retail Price in $ per month[27]
Bsp and Bsp^2 = Speed choice modelling attribute values
Bpr and Bpr^2 = Retail Price choice modelling attributes value

Fig. A.3 presents the Vertigan Panel's total direct annual benefits under each roll-out strategy. It can be demonstrated that the MWTP error identified at the household level cascades through to the Vertigan Panel's overall reported benefits. In the year 2022, for example, Fig. A.3 reports total benefits for the MTM roll-out strategy in fixed line areas of about $5 billion. This is roughly equivalent to the Vertigan Panel's assumed 10 million premises, adjusted for the 4.4 percent already rolled-out to in 2014, times an estimated 83 percent household broadband take-up, at $53 per premise per month for 12 months, with the $53 per premise being close to the Vertigan Panel's reported benefit of $46.50 monthly per household in 2014 uplifted for the Vertigan Panel's assumed 2 percent growth in the WTP per annum.[28] Whilst this analysis, which is reliant on reading data from the Vertigan Panel's reported charts, can only be indicative, it reveals that the Vertigan Panel's overall benefit results relied on the erroneous estimates of the WTP for broadband. If the recalculated WTP estimates in this study had been used, the 2022 benefits under the MTM roll-out strategy would have been about half of the amounts reported by the Vertigan Panel. Further, similar steps can be taken to demonstrate that the overstated annual benefits in Fig. A.3 cascade through to the Vertigan Panel's reported total NPV benefits for each rollout strategy.



Data source: The CIE.

Fig. A3. Vertigan Panel (2014) – Timing of Direct Benefits to Households and Businesses by Rollout Strategy.
Source: Extract from Vertigan et al. (2014b), Chart 6.2, p.60.

Appendix B. Recalculation of Vertigan Panel's Actual Take-up Method Results

Table B.1 recalculates the Vertigan Panel's actual take-up method benefits, after excluding benefits associated with the broadband speeds that were already available to households in 2014. Table B.2 reports a revised benefit per household that is 48 percent lower than that published by the Vertigan Panel.

Table B.1
Vertigan Panel (2014) – Recalculated Results, Actual Take-up Method, Incremental NBN Benefits per Household per Month, Fixed Line Footprint, MTM Roll-out, 2014 Dollars.

Speed Plan DL/UL (Mbps)	Vertigan Panel (2014) Inputs			Revised Benefit Calculations (2014)						
	Average Retail Price in 2014*	Share of Fixed Line NBN HHs subscribing to DL speed in 2014*	Share of Fixed Line NBN HH with access to DL speed in 2014**	Share of HH subscribing to DL speed or higher in 2014 [A]	Share of HH without access to DL speed in 2014 [B]	Share of HH with access to DL speed under MTM roll-out* [C]	Incremental Producer Surplus of DL speed	Incremental Consumer Surplus of DL speed**	Total incremental Surplus of DL speed [D]	Total Surplus contribution of DL speed [A x B x C x D]
12/1	$65.70	42.2%	41.2%	100.0%	58.8%	100.0%	$0.00	$8.40	$8.40	$4.94
25/5	$74.10	29.2%	28.3%	57.8%	71.7%	100.0%	$8.40	$2.10	$10.50	$4.35
25/10	$75.10	0.8%	28.3%	28.6%	71.7%	100.0%	$1.00	$0.25	$1.25	$0.26

(continued on next page)

[26] Vertigan Panel (2014b), p.182.
[27] The Vertigan Panel calculated the MWTP using a retail price of $75 monthly. Source: Vertigan Panel (2014b), pp.63-63.
[28] The Vertigan Panel (2014b) assumed 10 million premises (Chart 4.3, p.45), 95.6 precent not already rolled out to by NBN (Table 6.7, p.65), 83 percent broadband take-up (p.69), 2 percent annual WTP growth (p.77), and 5 percent WTP premium for the 11 percent of business premises (p.77). Calculation: $5b in 2022 = 10m premises x 95.6 percent not rolled out to x 83 percent take-up x 12 months x $53 per month.

Table B.1 (*continued*)

Speed Plan DL/UL (Mbps)	Vertigan Panel (2014) Inputs			Revised Benefit Calculations (2014)						
	Average Retail Price in 2014^	Share of Fixed Line NBN HHs subscribing to DL speed in 2014^	Share of Fixed Line NBN HH with access to DL speed in 2014^^	Share of HH subscribing to DL speed or higher in 2014 [A]	Share of HH without access to DL speed in 2014 [B]	Share of HH with access to DL speed under MTM roll-out* [C]	Incremental Producer Surplus of DL speed	Incremental Consumer Surplus of DL speed**	Total incremental Surplus of DL speed [D]	Total Surplus contribution of DL speed [A x B x C x D]
50/20	$82.10	5.0%	27.8%	27.8%	72.2%	100.0%	$7.00	$1.75	$8.75	$1.76
100/40	$91.70	22.8%	4.3%	22.8%	95.7%	0.0%	$9.60	$2.40	$12.00	$0.00
Total	$74.98									$11.30

Benefits incremental to the "No further roll-out" strategy. DL = download speed, UL = upload speed, HH = Households.

^ Source: Vertigan et al. (2014b), Table B.14, p.109. Total is weighted average using "Share of Fixed NBN subscribing to DL speed in 2014″ column.

^^ Source: Vertigan et al. (2014b), Table B.5, p.101. Due to need to read data from the Vertigan Panel charts, recalculations are indicative.

* Source: Vertigan et al. (2014b), p.75. Note the Vertigan Panel assumed MTM households would have access to all broadband speed tiers except the 100/40 tier.

** Calculated using Vertigan Panel assumptions, including benefit of $8.40 from moving to the 12/1 plan and price elasticity of demand of −2.0. Source: Vertigan et al. (2014b), pp.71–75.

Note: Only the MTM fixed line roll-out recalculation is shown above. The same recalculation is performed for other roll-out strategies, results presented in Table B.2.

Table B.2
Vertigan Panel (2014) - Published and Recalculated Results, Actual Take-up Method, Incremental NBN Benefits per Household per Month, Fixed Line Footprint, 2014 Dollars.

Roll-out Strategy	Vertigan Panel (2014) Published Results^			Vertigan Panel (2014) Recalculated Results^^			
	DL	UL	Total	DL	UL	Total	Total% Change
Unsubsidised Rollout	–	–	$21.80	–	–	$11.30	(48%)
MTM Rollout	–	–	$21.80	–	–	$11.30	(48%)
FTTP Rollout	–	–	$26.30	–	–	$13.92	(47%)

Benefits per month incremental to the "No further roll-out" strategy. DL = download speed, UL = upload speed. Note that actual take-up method did not breakdown WTP into DL and UL speeds.

^ Published results based on Table 1.

^^ Recalculated results based on Table B.1.

Appendix C. Discussion of Vertigan Panel's Time Savings Method

Fig. C.1 presents the time savings benefits per household reported by the Vertigan Panel. As shown, the benefits were forecast to be highly concentrated in those households that had access to the slowest available speeds in 2014. For households with speeds of less than about 10–15 Mbps, benefits were as high as about $100 per month. For households with speeds of more than about 10–15 Mbps, benefits were about $1 or less per month. As the time savings method does not consider income constraints, it is reasonable to expect that the benefits generated using the time savings method would have been materially higher than the benefits generated using the choice modelling method (which is designed to consider income constraints). In the Vertigan Panel's published results, however, the average household benefits generated by the time savings method ($53.21 per household per month under the MTM roll-out strategy) were similar to those generated using the choice modelling method ($46.50).

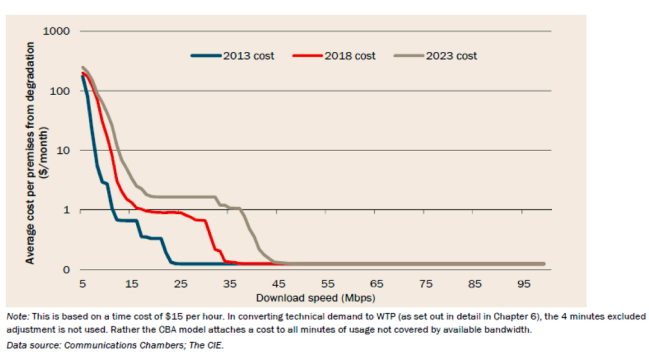


Fig C.1. Vertigan Panel (2014) – Time Savings Method, Cost of Degraded Download Minutes (log scale), 2014 Dollars.
Extract from Vertigan et al. (2014b), Chart 6.9, p.68.

Note that at the time of the Vertigan Panel release some industry analysts were critical of the Vertigan Panel's bandwidth demand estimates, claiming they were too conservative (Gregory 2014; Tucker 2014). These criticisms were primarily based on a comparison of the estimates relied on by the Vertigan Panel with the average speed results being reported by popular speed testing services, such as Ookla. Whereas the Vertigan Panel focused on the broadband speeds that would be *required* by households, speed testing services report the *actual* speeds available to the (often tech savvy) users that perform the tests, and it has been argued that relying on speed testing results to ascertain required speeds can be considered problematic (AlphaBeta, 2019). Whilst there is limited public information available on household bandwidth requirements, a comparison of the Vertigan Panel's bandwidth demand estimates with those more recently published in a comprehensive Australian government-sponsored study provides some evidence

of the Vertigan Panel's estimates being resilient over time (BCAR, 2020). Although the comparison indicates some relative over-estimation in the Vertigan Panel's bandwidth estimates for 2018 (at the 95th and 99th percentiles) and some relative under-estimation for 2023 (at the 50th and 99th percentiles), the reasonable degree of proximity in the estimates indicates that the future bandwidth requirements relied on by the Vertigan Panel were not systematically understated (see Table C.1).

Table C.1
Vertigan Panel (2014) vs. BCAR (2020) – Households Bandwidth Demand Forecasts (Mbps).

Percentile	Forecast for 2018			Forecast for 2023		
	Vertigan Panel (2014)	BCAR (2020)	Difference (Vertigan - BCAR)	Vertigan Panel (2014)	BCAR (2020)	Difference (Vertigan - BCAR)
50th percentile	12	14	−2	15	22	−7
95th percentile	33^	24	9	43	43	0
99th percentile	35	29	6	45	54	−9

^ Bandwidth requirement read from Kenny et al. (2014), Figure 24. All other bandwidth requirements extracted from tables or text. Comparable Vertigan Panel and BCAR data only available for the years 2018 and 2023.

Source: BCAR (2020), Vertigan Panel (2014) estimates sourced from Kenny et al. (2014), p.54–55, a report commissioned for the Vertigan Panel.

Appendix D. Choice Survey Artefacts

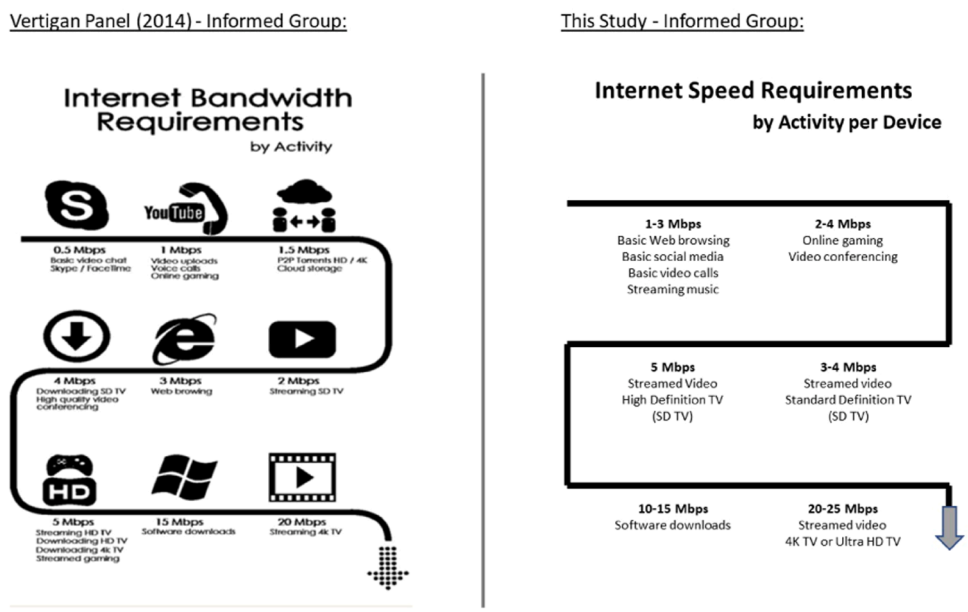
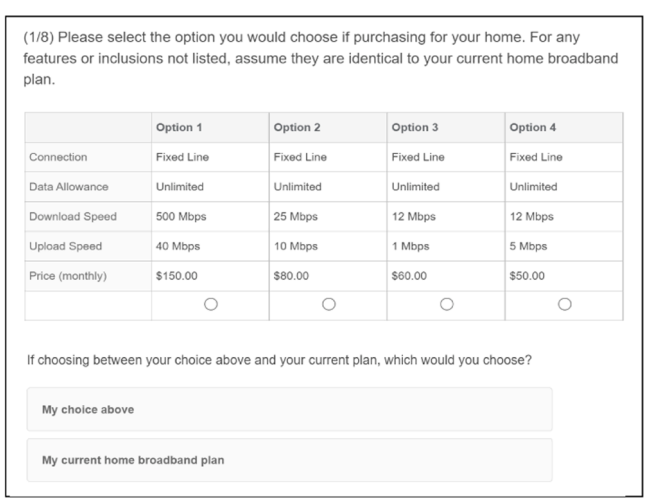


Fig. D.1. Vertigan Panel (2014) vs. This Study (2022) - Informed Group Information Aid.
Source: Vertigan Panel (2014), Volume 2, Chart H.1, p.167.



(1/8) Please select the option you would choose if purchasing for your home. For any features or inclusions not listed, assume they are identical to your current home broadband plan.

	Option 1	Option 2	Option 3	Option 4
Connection	Fixed Line	Fixed Line	Fixed Line	Fixed Line
Data Allowance	Unlimited	Unlimited	Unlimited	Unlimited
Download Speed	500 Mbps	25 Mbps	12 Mbps	12 Mbps
Upload Speed	40 Mbps	10 Mbps	1 Mbps	5 Mbps
Price (monthly)	$150.00	$80.00	$60.00	$50.00
	○	○	○	○

If choosing between your choice above and your current plan, which would you choose?

My choice above

My current home broadband plan

Fig. D.2. This Study (2022) - Choice Experiment Example Task.

Appendix E. Choice Survey Statistics

Table E.1
This Study (2022) - Demographic Characteristics: Survey Respondents vs. Australian Population.

Demographic Data	Uninformed Group	Informed Group	Survey Total	Australia
Sample Size	338	330	668	NA
	50.6%	49.4%	100.0%	NA
State				
New South Wales	35.2%	28.8%	32.0%	31.8%
Victoria	29.3%	29.4%	29.3%	25.8%
Queensland	14.8%	22.4%	18.6%	20.4%
Western Australia	8.3%	4.8%	6.6%	10.4%
South Australia	8.3%	7.6%	7.9%	6.9%
Tasmania	2.1%	2.4%	2.2%	2.1%
Australian Capital Territory	1.8%	3.3%	2.5%	1.7%
Northern Territory	0.3%	1.2%	0.7%	1.0%
Region				
Major Cities of Australia	82.2%	77.9%	80.1%	72.2%
Inner Regional Australia	13.3%	14.8%	14.1%	17.9%
Outer Regional Australia	3.3%	5.8%	4.5%	8.0%
Remote Australia	0.3%	0.3%	0.3%	1.1%
Very Remote Australia	0.0%	0.0%	0.0%	0.8%
Other	0.9%	1.2%	1.0%	0.0%
Age (years)				
18–24	3.9%	4.8%	4.3%	11.0%
25–34	17.0%	16.7%	16.9%	18.8%
35–44	24.8%	22.4%	23.6%	17.5%
45–54	19.3%	17.0%	18.1%	16.2%
55–64	14.1%	18.5%	16.3%	14.9%
65 or older	20.9%	20.6%	20.8%	21.6%
Gender				
Male	46.9%	49.4%	48.2%	49.0%
Female	52.7%	50.6%	51.7%	51.0%
Non-Binary	0.3%	0.0%	0.2%	0.0%
Employment				
Employed	66.6%	64.8%	65.7%	63.8%
Unemployed	3.6%	2.4%	3.0%	2.6%
Other	29.9%	32.7%	31.3%	33.6%
Occupation				
Professionals	26.2%	28.4%	27.3%	22.9%
Managers	24.9%	21.2%	23.1%	13.3%
Clerical and Administrative	16.0%	19.4%	17.7%	13.7%
Technicians and Trades	5.5%	6.8%	6.1%	12.4%
Community and Personal Service	3.0%	4.1%	3.5%	12.1%
Labourers	8.4%	7.2%	7.8%	10.7%
Sales workers	6.3%	5.0%	5.7%	8.8%
Machinery Operators and Drivers	0.0%	2.7%	1.3%	6.0%
Other	9.7%	5.4%	7.6%	0.0%
Household Income (gross, weekly)				
<$1000 per week	27.5%	30.0%	28.7%	27.4%
$1000-$1999 per week	29.3%	34.2%	31.7%	27.9%
$2000-$2999 per week	24.3%	20.6%	22.5%	19.1%
$3000-$3999 per week	8.3%	7.9%	8.1%	12.0%
$4000 or more per week	10.7%	7.3%	9.0%	13.7%
Household Composition				
Average size	2.8	2.7	2.7	2.6
Lone person	16.6%	19.4%	18.0%	25.2%
Couple only	29.3%	32.7%	31.0%	26.3%
One parent / Couple with children	33.7%	30.9%	32.3%	37.5%
Other	20.4%	17.0%	18.7%	10.9%

Australian population data from multiple Australian Bureau of Statistics sources, available on request. Note that ABS household data defines "child" to include a dependant student aged 18–24. In this study, child only includes dependents under age 18. For this reason, the "Other" household composition category is larger in this study, and includes households with three or four adults, some of whom could be dependant students.

Table E.2
This Study (2022) – Broadband Subscription Characteristics: Survey Respondents vs. Australian Population.

Broadband Subscription Data	Uninformed Group	Informed Group	Survey Total	Australia
Home Broadband - Access Technology				
Fixed Broadband	70.4%	73.6%	72.0%	NA^
Home Wireless Broadband	29.6%	26.4%	28.0%	NA^

(*continued on next page*)

Table E.2 (*continued*)

Broadband Subscription Data	Uninformed Group	Informed Group	Survey Total	Australia
Fixed Broadband - Access Technology				
Fixed Line	84.0%	82.7%	83.4%	94.3%
Fixed Wireless	14.7%	10.7%	12.7%	4.5%
Satellite	1.3%	6.6%	3.9%	1.3%
Fixed Broadband - Access Provider				
NBN Co	92.4%	90.1%	91.3%	NA^
Fixed Broadband - Retail Service Provider				
Telstra*	33.2%	35.0%	34.1%	43.7%
TPG	19.3%	20.6%	19.9%	23.3%
Optus	14.3%	15.2%	14.8%	13.9%
Vocus	8.0%	6.6%	7.3%	7.3%
Aussie Broadband	8.0%	8.6%	8.3%	6.1%
Superloop	0.8%	0.8%	0.8%	1.8%
Other*	13.0%	10.7%	11.9%	3.8%
Vodafone	3.4%	2.5%	2.9%	NA^
Fixed Broadband - Download Speed				
Fixed Line - 12 Mbps	8.8%	5.3%	7.1%	9.6%
Fixed Line - 25 Mbps	16.4%	15.6%	16.0%	11.4%
Fixed Line - 50 Mbps	29.0%	32.1%	30.5%	55.3%
Fixed Line - 100 Mbps	13.9%	16.0%	14.9%	15.6%
Fixed Line - 250 Mbps	6.7%	7.4%	7.1%	1.9%
Fixed Line - 500–1000 Mbps	9.2%	6.2%	7.7%	0.5%
Fixed Wireless	14.7%	10.7%	12.7%	4.5%
Satellite	1.3%	6.6%	3.9%	1.3%
Fixed Broadband - Data Allowance				
Unlimited Data	84.0%	84.0%	84.0%	80.0%
Limited Data	16.0%	16.0%	16.0%	20.0%
Home Broadband – Payments				
Average Monthly Payments	$94	$95	$95	NA^

Fixed Broadband includes fixed line broadband, fixed wireless broadband and satellite broadband.

^ Data source for the Australian population not identified.

* Telstra provides services under the brands "Telstra" and "Belong". The survey did not separately capture the Belong brand. It is expected that Belong survey respondents will have selected "Other", although they are reported as part of Telstra in the Australia totals.

Source for Australia data: ACCC (2022).

Table E.3
This Study (2022) – Survey Respondents, Status Quo Selection.

Status Quo Question	Uninformed Group	Informed Group	Survey Total
Status Quo Preference			
Current Home Broadband Plan	54.9%	53.7%	54.3%
Hypothetical Choice	45.1%	46.3%	45.7%

Appendix F Choice Survey Comparison

Table F.1
This Study (2022) - Vertigan Panel (2014) Attributes Removed or Held Constant, 2014 Dollars.

Vertigan Panel (2014) Attributes				This Study (2022) Attribute Approach and Reasons	
Attribute Included	Attribute levels	Average attribute WTP per month^	Average attribute market share impact^^	Attribute approach	Reasons for approach (comments refer to the Australian market)
Data allowance	20–1000 GB per month	NA	32%	Included, but held constant	All identified broadband plans have unlimited data in 2022.
Data peak /off peak restrictions	No restrictions, Limits apply	$1.89 (for no restrictions)	2%	Excluded	All identified broadband plans have no peak/off-peak restrictions in 2022.
Pay TV service Availability	Available, (additional costs apply) Not available	$0.77 (for PayTV available)	1%	Excluded	All major fixed broadband operators offer either PayTV or streaming services as optional add-ons in 2022. Further, households can readily access over-the-top services (e.g., Netflix). Hence, the attribute is not considered to be a choice differentiator.
Mobile service Availability	Available, (additional costs apply) Not available	$1.19 (for mobile available)	2%	Excluded	All major fixed broadband operators offer mobile services in 2022. Hence, the attribute is note considered to be a choice differentiator.
Home Phone inclusion	Included, Not included	$7.02 (for home phone included)	10%	Excluded	Home phone options are not prominently promoted by retail service providers in 2022. Further, home phone take-up has declined substantially.

(*continued on next page*)

Table F.1 (*continued*)

Vertigan Panel (2014) Attributes				This Study (2022) Attribute Approach and Reasons	
Attribute Included	Attribute levels	Average attribute WTP per month^	Average attribute market share impact^^	Attribute approach	Reasons for approach (comments refer to the Australian market)
WiFi Modem inclusion	Included, Not included	$0.66 (for WiFi included)	1%	Excluded	Where a Wi-Fi modem is included, it is usually accompanied by higher set-up fees, a contract period, or equipment pay-out fee in 2022. Hence, the attribute is confounded with other attributes.
Contact Term	No contract, Contract (12 or 24 months)	$1.69 (for no contract)	2%	Excluded	Most plans are offered under either month-to-month (MTM) or contract terms in 2022. Where MTM plans are offered, they are generally accompanied by higher set-up costs or equipment/ modem pay-out arrangements. Hence, the attribute is likely to be confounded with other attributes.
Brand	Nine brand alternatives	NA	NA	Excluded	Although brand alternatives were included in the Vertigan Panel design, brand market share impacts were not shown in the results.

^ Average WTP calculated using Vertigan Panel attribute values. Source: Vertigan et al. (2014b), Tables H.7 and H.8, pp.178–179.
^^ Average market share impact calculated as average of informed and uninformed respondent groups. Source: Vertigan et al. (2014b), Table H.9, p.181.

Appendix G. Discussion of the Vertigan Panel's International Research Analysis

The Vertigan Panel reviewed the then available international literature on the WTP for broadband and identified two studies that it considered to be particularly relevant, being Dutz et al. (2009) and Rosston et al. (2010).[29] Dutz et al. (2009) had found that households were willing to pay US$7.20 (AUD $7.60 in 2014) per month to move from download speeds of 5 Mbps to speeds of 50 Mbps. In contrast, the Vertigan Panel published results indicated that households were willing to pay $47.00 per month for similar speed increases.[30] Despite the sizeable difference in WTP estimates, the Vertigan Panel did not explore the difference and appears to have overlooked Dutz et al. (2009) on erroneous grounds.[31] Rosston et al. (2010) had found that US households were willing to pay just US$3.00 (AUD $3.20 in 2014) per month to move from "fast" to "very fast" broadband speeds.[32] The Vertigan Panel, however, doubted the credibility of the Rosston et al. (2010) study, also on erroneous grounds, and did not explore the sizeable estimation differences.[33] Hence, whilst both Dutz et al. (2009) and Rosston et al. (2010) had provided evidence that the household MWTP for faster broadband speeds was low, potentially in the range of $3–8 per month for significant speed increases, the findings were not given serious consideration by the Vertigan Panel. Further, although both studies identified significant heterogeneity of demand, the impact of heterogeneity on future NBN penetration rates was not considered by the Vertigan Panel.

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[29] Source: Vertigan Panel (2014b), Appendix D, pp.115. Note that US dollar conversions to AUD in this section are those reported by the Vertigan Panel.

[30] Calculated from the Vertigan Panel's choice model charts and attribute values, based on download speeds. Source: Vertigan Panel (2014b), Charts 6.4 and 6.5, pp.63-4, Tables H.7 and H.8, pp.178-179.

[31] Dutz et al. (2009) used a simple preference instrument to determine WTP and the research was conducted four years prior to the Vertigan Panel study. The Vertigan Panel, however, made incorrect claims regarding Dutz et al. (2009). First, the Vertigan Panel stated that Dutz et al. (2009) included "no reported use of video", whereas Dutz et al. (2009) reported that in 2008, 21.3 percent of US broadband users were using video (pp.14-15). Second, the Vertigan Panel stated that the Dutz et al. (2009) estimates were based on the sample period from 2005 to 2008. However, the Dutz el al. (2009) estimates of the WTP for increasing broadband speeds from 5 Mbps to 50 Mbps were based on a survey conducted in March 2009 (see p.26). Third, the Vertigan Panel stated that Dutz et al. (2009) was different because "only 69 percent of the sample had a broadband connection … and 20 percent still had no internet connection". However, ABS data relied upon by the Vertigan Panel in 2013 reported that 83 percent of Australian households used the internet and 77 percent of those households had a broadband connection, which was not materially different to the US situation outlined in Dutz et al. (2009). Fourth, the Vertigan Panel stated that a limitation of Dutz et al. (2009), when trying to apply it to Australia, was that there was "no wireless broadband option (again in sharp contrast to Australian usage)". However, Dutz et al. (2009) states that the "sizeable benefits to households from mobile wireless broadband" (see p.5) are additional to their estimates, hence taking a similar approach to wireless broadband benefits as that taken by the Vertigan Panel, which also excluded wireless broadband from its benefit estimates. See Vertigan Panel (2014b), p.155.

[32] Rosston et al. (2010) used the following descriptions with survey respondents: Fast: "Much faster downloads and uploads. It is great for music, photo sharing, and watching some videos"; Very Fast: "Blazing fast downloads and uploads. It is really great for gaming, watching high-definition movies, and instantly transferring large files."

[33] The Vertigan Panel questioned the credibility of Rosston et al. (2010) on the basis that the US Office of Management and Budget (OMB) had found that the study did not meet its guidance to be considered nationally representative. However, Rosston et al. (2010) was only audited by the OMB because it was it was funded by the FCC. Further, although Rosston et al. (2010) did not include any analysis of survey non-respondents, which was the basis for the OMB finding, it did include a detailed demographic comparison of the survey respondents with the US population, demonstrating a close alignment across most demographic characteristics. A similar demographic comparison was not provided by the Vertigan Panel.

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